

- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients
- Not recommended for use in children under age 12
- Several studies show lack of efficacy of TCAs for depression
- May be used to treat enuresis or hyperactive/impulsive behaviors
- Some cases of sudden death have occurred in children taking TCAs
- Adolescents: initial dose 50 mg/day; increase gradually up to 100 mg/day

### **Pregnancy**

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women

- Crosses the placenta
- Adverse effects have been reported in infants whose mothers took a TCA (lethargy, withdrawal symptoms, fetal malformations)
- Must weigh the risk of treatment (first trimester fetal development, third trimester newborn delivery) to the child against the risk of no treatment (recurrence of depression, maternal health, infant bonding) to the mother and child
- For many patients this may mean continuing treatment during pregnancy
- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

### **Breast Feeding**

- Some drug is found in mother's breast milk
- \* Recommended either to discontinue drug or bottle feed
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Must weigh the risk of treatment (first trimester fetal development, third trimester newborn delivery) to the child against the risk of no treatment (recurrence of depression, maternal health, infant bonding) to the mother and child

- For many patients this may mean continuing treatment during breast feeding

## **The Art Of Psychopharmacology**

### **Potential Advantages**

- Patients with insomnia, anxiety
- Severe or treatment-resistant depression

### **Potential Disadvantages**

- Pediatric and geriatric patients
- Patients concerned with weight gain and sedation

### **Primary Target Symptoms**

- Depressed mood
- Symptoms of anxiety
- Somatic symptoms

### **Pearls**

- \* May be more useful than some other TCAs for patients with anxiety, sleep disturbance, and depression with physical illness
- \* May be more sedating than some other TCAs
- TCAs are often a first-line treatment option for chronic pain

- TCAs are no longer generally considered a first-line option for depression because of their side effect profile
- TCAs continue to be useful for severe or treatment-resistant depression
- TCAs may aggravate psychotic symptoms
- Alcohol should be avoided because of additive CNS effects
- Underweight patients may be more susceptible to adverse cardiovascular effects
- Children, patients with inadequate hydration, and patients with cardiac disease may be more susceptible to TCA-induced cardiotoxicity than healthy adults
- For the expert only: although generally prohibited, a heroic but potentially dangerous treatment for severely treatment-resistant patients is for an expert to give a tricyclic/tetracyclic antidepressant other than clomipramine simultaneously with an MAOI for patients who fail to respond to numerous other antidepressants
- If this option is elected, start the MAOI with the tricyclic/tetracyclic antidepressant simultaneously at low doses after appropriate drug washout, then alternately increase doses of these agents every few days to a week as tolerated
- Although very strict dietary and concomitant drug restrictions must be observed to prevent hypertensive crises and serotonin syndrome, the most common side effects of MAOI and tricyclic/tetracyclic

antidepressant combinations may be weight gain and orthostatic hypotension

- Patients on tricyclics should be aware that they may experience symptoms such as photosensitivity or blue-green urine
- SSRIs may be more effective than TCAs in women, and TCAs may be more effective than SSRIs in men
- Since tricyclic/tetracyclic antidepressants are substrates for CYP450 2D6, and 7% of the population (especially Caucasians) may have a genetic variant leading to reduced activity of 2D6, such patients may not safely tolerate normal doses of tricyclic/tetracyclic antidepressants and may require dose reduction
- Phenotypic testing may be necessary to detect this genetic variant prior to dosing with a tricyclic/tetracyclic antidepressant, especially in vulnerable populations such as children, elderly, cardiac populations, and those on concomitant medications
- Patients who seem to have extraordinarily severe side effects at normal or low doses may have this phenotypic CYP450 2D6 variant and require low doses or switching to another antidepressant not metabolized by 2D6

### *Suggested Reading*

Anderson IM . Selective serotonin reuptake inhibitors versus tricyclic antidepressants: a meta-analysis of efficacy and tolerability . *J Aff Disorders*

2000 ;**58** :19 –36 .

Anderson IM . Meta-analytical studies on new antidepressants . *Br Med Bull* 2001 ;**57** :161–78 .

Berger M , Gastpar M . Trimipramine: a challenge to current concepts on antidepressives . *Eur Arch Psychiatry Clin Neurosci* 1996 ;**246** :235–9 .

Lapierre YD . A review of trimipramine. 30 years of clinical use . *Drugs* 1989 ;**38** (Suppl 1):S17 –24; discussion S49 –50.

Valbenazine

## Therapeutics

### Brands

- Ingrezza
- *see index for additional brand names*

### Generic?

No

### Class

- Vesicular monoamine transporter 2 (VMAT2) inhibitor

### Commonly Prescribed for

*(bold for FDA approved)*

- **Tardive dyskinesia in adults**

### How the Drug Works

- Valbenazine is a selective and reversible inhibitor of the vesicular monoamine transporter 2 (VMAT2), which packages monoamines, including dopamine, into synaptic vesicles of presynaptic neurons in the central nervous system

### **How Long Until It Works**

- In clinical trials valbenazine separated from placebo as early as week 2

### **If It Works**

- Patients should experience a significant reduction in the total Abnormal Involuntary Movement Scale (AIMS) score

### **If It Doesn't Work**

If tardive dyskinesia does not reverse with valbenazine, then other management options include deutetrabenazine, tetrabenazine, reserpine, clonazepam, amantadine, botulinum toxin injections for focal dystonia symptoms, or ginkgo biloba

Some patients may require suppressive therapy, in which the offending antipsychotic is reinstituted or its dose raised; this should only be considered if the symptoms are very severe and cause marked functional impact (e.g., inability to eat; mouth sores from rubbing); this can make tardive dyskinesia worse in the long run, but the short-term benefits may justify the risk for *certain* patients



### **Best Augmenting Combos for Partial Response or Treatment Resistance**

- Valbenazine is itself an augmenting agent to treat a side effect of antipsychotic drugs



## Tests

None for healthy individuals; no need for CYP450 2D6 testing

## Side Effects

### How Drug Causes Side Effects

- Theoretically due to increases in monoamine concentrations at receptors in parts of the brain and body other than those that cause therapeutic actions
- Depletion of dopamine due to long-term inhibition of VMAT2 may also be responsible for some side effects

### Notable Side Effects

Sedation

### Life-Threatening or Dangerous Side Effects

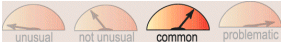
- QT prolongation, although the degree of QT prolongation is not clinically significant at concentrations expected with recommended dosing

### Weight Gain



- Reported but not expected

## Sedation



- Many experience and/or can be significant in amount

## What to Do About Side Effects

- Wait
- Wait
- Wait

## Best Augmenting Agents for Side Effects

Valbenazine is itself an augmenting agent to treat a side effect of antipsychotic drugs

## Dosing And Use

### Usual Dosage Range

- 40–80 mg once daily

### Dosage Forms

- Capsule 40 mg

### How to Dose

- Initial dose 40 mg once daily; after 1 week increase to 80 mg once daily

## **Dosing Tips**

Pharmacokinetics permit once daily dosing

Can be taken with or without food

Consider taking at night to reduce daytime sedation

If improvements in movements wear off before the end of the day with nighttime dosing, consider taking in the morning

## **Overdose**

- Limited experience

## **Long-Term Use**

- Clinical trials have shown continued improvement through 48 weeks

## **Habit Forming**

- No

## **How to Stop**

- Tapering not necessary
- In clinical trials, tardive dyskinesia symptoms returned during the 4-week period following discontinuation of valbenazine

## **Pharmacokinetics**

- Valbenazine is extensively metabolized by hydrolysis of the valine ester to form the active metabolite ([+]- $\hat{I}$ -HTBZ) and by oxidative metabolism, primarily by CYP450 3A4/5, to form monooxidized valbenazine and other minor metabolites
- [+]- $\hat{I}$ -HTBZ is further metabolized in part by CYP450 2D6
- Valbenazine and [+]- $\hat{I}$ -HTBZ have half-lives of 15–22 hours
- Very little of the parent drug, valbenazine, is present in the blood due to rapid conversion to its metabolites

### **Drug Interactions**

Strong CYP450 3A4 inducers (e.g., carbamazepine) may reduce exposure to valbenazine; use with strong CYP3A4 inducers is not recommended

Strong CYP450 3A4 inhibitors (e.g., ketoconazole) may increase exposure to valbenazine; reduce dose to 40 mg in patients taking a strong CYP3A4 inhibitor

Strong CYP450 2D6 inhibitors (e.g., paroxetine, fluoxetine) may increase exposure to valbenazine's active metabolite; consider reducing dose based on tolerability in patients taking a strong CYP450 2D6 inhibitor or who are known CYP2D6 poor metabolizers

Not recommended for use with monoamine oxidase inhibitors (MAOIs)

Valbenazine may increase concentrations of digoxin, so monitor digoxin concentrations in patients taking concomitant digoxin; dose

adjustment may not be necessary

### **Other Warnings/Precautions**

May cause an increase in QT interval, although the degree of QT prolongation is not clinically significant at concentrations expected with recommended dosing

Avoid use in patients with congenital long QT syndrome or with arrhythmias associated with a prolonged QT interval

For patients at increased risk of prolonged QT interval, assess the QT interval before increasing the dose

### **Do Not Use**

- If there is a proven allergy to valbenazine

## **Special Populations**

### **Renal Impairment**

- Mild to moderate impairment: dose adjustment not necessary
- Severe impairment: not recommended for use

### **Hepatic Impairment**

- Moderate to severe impairment: 40 mg once daily

### **Cardiac Impairment**

- May cause an increase in QT interval; avoid use in patients with congenital long QT syndrome or with arrhythmias associated with a prolonged QT interval

### **Elderly**

- Dose adjustment not necessary

### **Children and Adolescents**

- Safety and efficacy have not been established

### **Pregnancy**

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- In animal studies, no malformations were observed when valbenazine was administered during the period of organogenesis at doses up to 1.8 (rat) or 24 (rabbit) times the maximum recommended human dose
- In rat studies, administration during organogenesis through lactation produced an increase in the number of stillborn pups and postnatal

pup mortalities at doses less than the maximum recommended human dose

### **Breast Feeding**

- Unknown if valbenazine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- Breast feeding is not recommended, including for 5 days after the final dose

## **The Art Of Psychopharmacology**

### **Potential Advantages**

- Once-a-day administration

### **Potential Disadvantages**

- Expensive
- Few dosing options for titration or individualizing dosing

### **Primary Target Symptoms**

- Repetitive involuntary movements of tardive dyskinesia (usually associated with lower facial and distal extremity musculature, e.g., tongue protrusion, writhing of tongue, lip smacking, chewing, blinking, and grimacing)

## Pearls

In clinical trials, valbenazine was well tolerated, with the frequency of adverse events similar in all treatment groups

Valbenazine improved tardive dyskinesia regardless of the therapeutic use or type of antipsychotic drug concomitantly administered

80 mg is generally associated with better efficacy than 40 mg

Valbenazine and its active metabolites bind predominantly only to VMAT2, so there are no significant off-target pharmacologic actions such as blocking the D2 dopamine receptor or the 5HT7 receptor, as is the case with tetrabenazine's and deutetabenazine's active metabolites

Can rarely cause akathisia and dry mouth

## *Suggested Reading*

Grigoriadis DE, Smith E, Hoare SR, Madan A, Bozigian H. Pharmacologic characterization of valbenazine (NBI-98854) and its metabolites. *J Pharmacol Exp Ther* 2017 ;361(3):454–61. doi: 10.1124/jpet.116.239160 .

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Hauser RA, Factor SA, Marder SR, et al. KINECT 3: a phase 3 randomized, double-blind, placebo-controlled trial of valbenazine for tardive dyskinesia. *Am J Psychiatry* 2017 ;174(5):476–84.

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Meyer JM. Forgotten but not gone: new developments in the understanding and treatment of tardive dyskinesia. *CNS Spectr* 2016 ;21(S1):13–24.



Valproate

## Therapeutics

### Brands

- Depakene
- Depacon
- Depakote, Depakote ER
- Stavzor
- *see index for additional brand names*

### Generic?

Yes

### Class

- Neuroscience-based Nomenclature: glutamate (Glu); yet to be determined
- Anticonvulsant, mood stabilizer, migraine prophylaxis, voltage-sensitive sodium channel modulator

### Commonly Prescribed for

*(bold for FDA approved)*

- **Acute mania (divalproex) and mixed episodes (divalproex, divalproex ER, valproic acid delayed-release)**
- **Complex partial seizures that occur either in isolation or in association with other types of seizures (monotherapy and adjunctive)**
- **Simple and complex absence seizures (monotherapy and adjunctive)**
- **Multiple seizure types, which include absence seizures (adjunctive)**
- **Migraine prophylaxis (divalproex, divalproex ER, valproic acid delayed-release)**
- Maintenance treatment of bipolar disorder
- Bipolar depression
- Psychosis, schizophrenia (adjunctive)

### **How the Drug Works**

- \* Blocks voltage-sensitive sodium channels by an unknown mechanism
- Increases brain concentrations of gamma-aminobutyric acid (GABA) by an unknown mechanism

### **How Long Until It Works**

- For acute mania, effects should occur within a few days depending on the formulation of the drug
- May take several weeks to months to optimize an effect on mood stabilization
- Should also reduce seizures and improve migraine within a few weeks

### **If It Works**

- The goal of treatment is complete remission of symptoms (e.g., mania, seizures, migraine)
- Continue treatment until all symptoms are gone or until improvement is stable and then continue treating indefinitely as long as improvement persists
- Continue treatment indefinitely to avoid recurrence of mania, depression, seizures, and headaches

### **If It Doesn't Work (for Bipolar Disorder)**

\* Many patients have only a partial response where some symptoms are improved but others persist or continue to wax and wane without stabilization of mood

- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Consider checking plasma drug level, increasing dose, switching to another agent, or adding an appropriate augmenting agent

- Consider adding psychotherapy
- Consider the presence of noncompliance and counsel patient
- Switch to another mood stabilizer with fewer side effects
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)

### **Best Augmenting Combos for Partial Response or Treatment Resistance (for Bipolar Disorder)**

- Lithium
- Atypical antipsychotics (especially risperidone, olanzapine, quetiapine, ziprasidone, and aripiprazole)
- \* Lamotrigine (with caution and at half the dose in the presence of valproate because valproate can double lamotrigine levels)
- \* Antidepressants (with caution because antidepressants can destabilize mood in some patients, including induction of rapid cycling or suicidal ideation; in particular consider bupropion; also SSRIs, SNRIs, others; generally avoid TCAs, MAOIs)

### **Tests**

- \* Before starting treatment, complete blood counts, coagulation tests, and liver function tests
- Consider coagulation tests prior to planned surgery or if there is a history of bleeding

- During the first few months of treatment, regular liver function tests and platelet counts; this can be shifted to once or twice a year for the remainder of treatment
- Plasma drug levels can assist monitoring of efficacy, side effects, and compliance
- \* Since valproate is frequently associated with weight gain, before starting treatment, weigh all patients and determine if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)
- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management
- \* Monitor weight and BMI during treatment
- \* While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different agent

## **Side Effects**

### **How Drug Causes Side Effects**

- CNS side effects theoretically due to excessive actions at voltage-sensitive sodium channels

### **Notable Side Effects**

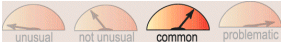
- \* Sedation, dose-dependent tremor, dizziness, ataxia, asthenia, headache
- \* Abdominal pain, nausea, vomiting, diarrhea, reduced appetite, constipation, dyspepsia, weight gain
- \* Alopecia (unusual)
- Polycystic ovaries (controversial)
- Hyperandrogenism, hyperinsulinemia, lipid dysregulation (controversial)
- Decreased bone mineral density (controversial)



### **Life-Threatening or Dangerous Side Effects**

- Can cause tachycardia or bradycardia
- Rare hepatotoxicity with liver failure sometimes severe and fatal, particularly in children under 2 years old
- Rare pancreatitis, sometimes fatal
- Rare but serious skin condition known as drug reaction with eosinophilia and systemic symptoms (DRESS)
- Rare activation of suicidal ideation and behavior (suicidality)

## Weight Gain



- Many experience and/or can be significant in amount
- Can become a health problem in some

## Sedation



- Frequent and can be significant in amount
- Some patients may not tolerate it
- Can wear off over time
- Can reemerge as dose increases and then wear off again over time

## What to Do About Side Effects

- Wait
- Wait
- Wait
- Take at night to reduce daytime sedation, especially with divalproex ER
- Lower the dose
- Switch to another agent

## Best Augmenting Agents for Side Effects

- \* Propranolol 20–30 mg 2–3 times/day may reduce tremor

\* Multivitamins fortified with zinc and selenium may help reduce alopecia

- Many side effects cannot be improved with an augmenting agent

## **Dosing And Use**

### **Usual Dosage Range**

- Mania: 1200–1500 mg/day
- Migraine: 500–1000 mg/day
- Epilepsy: 10–60 mg/kg per day

### **Dosage Forms**

- Tablet [delayed-release, as divalproex sodium (Depakote)] 125 mg, 250 mg, 500 mg
- Tablet [extended-release, as divalproex sodium (Depakote ER)] 250 mg, 500 mg
- Capsule [sprinkle, as divalproex sodium (Depakote Sprinkle)] 125 mg
- Capsule [as valproic acid (Depakene)] 250 mg
- Capsule [delayed-release, as valproic acid (Stavzor)] 125 mg, 250 mg, 500 mg
- Injection [as sodium valproate (Depacon)] 100 mg/mL (5 mL)



## How to Dose

- Usual starting dose for mania or epilepsy is 15 mg/kg in 2 divided doses (once daily for extended-release valproate)
- Acute mania (adults): initial 1000 mg/day; increase dose rapidly; maximum dose generally 60 mg/kg per day
- For less acute mania, may begin at 250–500 mg the first day, and then titrate upward as tolerated
- Migraine (adults): initial 500 mg/day; maximum recommended dose 1000 mg/day
- Epilepsy (adults): initial 10–15 mg/kg per day; increase by 5–10 mg/kg per week; maximum dose generally 60 mg/kg per day



## Dosing Tips

- \* Oral loading with 20–30 mg/kg per day may reduce onset of action to 5 days or less and may be especially useful for treatment of acute mania in inpatient settings
- Given the half-life of immediate-release valproate (e.g., Depakene, Depakote), twice daily dosing is probably ideal
- Extended-release valproate (e.g., Depakote ER) can be given once daily
- However, extended-release valproate is only about 80% as bioavailable as immediate-release valproate, producing plasma drug levels 10–20% lower than with immediate-release valproate